Lexical exceptionality in paradigm-specific learning: modeling stem-final obstruent alternations in Korean verbs and adjectives

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Abstract

Korean stem-final conjugations illustrate the interaction between lexical exceptionality and heterogeneous phonological processes. When /p/-, /t/-, and /s/-final stems occur before vowel-initial suffixes, the irregular classes in these paradigms undergo intervocalic lenition, each adopting a distinct alternation strategy. This study explores how learners acquire lexically-specific phonological patterns given multiple possible solutions. Specifically, it employs a lexically scaled MaxEnt model (Hughto et al., 2019; Linzen et al., 2013) to learn these paradigm-specific alternations. We run simulations under two learning scenarios: (1) when repair strategies occur at equal frequencies and (2) when one strategy significantly outnumbers the others. Results show that the model favors a *least-cost* solution by treating statistically dominant classes of morphemes as the general pattern. We conclude by discussing the model's sensitivity to lexical statistics, predictions for empirical testing, as well as its implications and challenges for language acquisition.

1 Introduction

Lexical exceptionality requires learners to reconcile two competing generalizations within a single grammatical system. When exceptional morphemes resist the general pattern, they challenge the constraint rankings that otherwise account for the majority of forms. Previous approaches to addressing such conflicts include cophonology (Kirchner, 1993; Inkelas et al., 1997; Anttila, 2002) and lexical indexation (Tesar and Smolensky, 1998; Tesar and Prince, 2003; Coetzee, 2009). Cophonology assumes separate rankings for different morphological classes, thereby permitting multiple subgrammars to resolve the conflict. Lexical indexation, by contrast, indexes constraints to specific lexical classes, allowing a single ranking to accommodate the conflict through lexically restricted constraint application.

While these approaches may suffice to account for exceptionality involving a single phonological process, they may lead to substantial ranking complexity when multiple processes are involved. Many languages employ more than one strategy to repair dispreferred phonological sequences (Zuraw and Lu, 2009). When multiple repair strategies intersect with lexical exceptionality, the learner must determine not only rankings for exceptional and non-exceptional items but also rankings among competing repairs. This raises the question of how learners reconcile inconsistency across two dimensions: lexical specificity and competing repair strategies.

Korean stem-final conjugations exemplify such empirical patterns and serve as a useful test case for this question. When irregular classes of verbs and adjectives ending in /p/, /t/, and /s/ combine with vowel-initial suffixes, the stem-final obstruents surface as [w], [ɾ], and ∅ through intervocalic lenition. The three lenition strategies serve the common goal of increasing sonority in the dispreferred *V-OBS-V sequence, but each strategy applies only to a specific stem-final paradigm.

To study this *paradigm-specific lexical exceptionality*, we adopt a lexically scaled MaxEnt framework (Hughto et al., 2019; Linzen et al., 2013). Unlike the two approaches discussed earlier, the model encodes both lexical exceptionality and competing repair strategies within a single set of weights. Lexical specificity is implemented through additive morpheme-specific constraint weights applied to particular paradigm classes, effectively capturing the role of indexed constraints while avoiding the need to posit separate subgrammars.

Moreover, the distribution of irregular forms varies across stem classes, allowing us to examine how statistical properties influence the choice of repair strategy. For stems ending in /p/, the majority exhibit irregular alternations, whereas stems

ending in /t/ and /s/ show roughly equal proportions of irregular and regular forms. To investigate the effect of this distributional asymmetry, we ran a toy simulation under two learning scenarios: (1) when only /t/- and /s/-final paradigms with roughly equal proportions of regular and irregular forms are considered, and (2) when the /p/-final paradigm, which exhibits a strong majority of irregular forms, is added. For both experimental conditions, the number of suffixes was incrementally increased to explore how the degree of affix exposure influence success in attributing lexical specificity to stems.

Results show improved generalization as the number of suffix types increases.[1] When given only a few suffix types, additive morpheme-specific weights were incorrectly assigned to suffixes. As the suffix inventory grew, these weights gradually shifted toward stems, correctly identifying them as the locus of lexical specificity. This process first involved differentiating regular and irregular classes and was further refined as the model began to parse finer distinctions among irregular classes. In the experiment including the /p/-final paradigm, the learner treated the /p/-irregular alternation as the default repair when lenition was triggered, as the constraint violated by this class showed the greatest resistance to receiving general weight. This configuration came at the cost of reduced prediction accuracy for the other two alternating classes.

The rest of the paper is organized as follows. Section 2 introduces the empirical patterns. Section 3 reviews the relevant literature and motivates the use of lexically scaled MaxEnt to model Korean stem-final conjugation. Section 4 presents the implementation details of the simulations, and Section 5 summarizes the results. Finally, Section 6 discusses the implications of these findings and outlines how learning may proceed when real-world data are fully incorporated.

2 Korean stem-final verb / adjective conjugations

Korean verbs and adjectives are traditionally divided into regular and irregular classes. Regularity is defined as invariance: regular forms remain unchanged throughout derivation, whereas irregular forms exhibit alternation before certain suffixes (Lee, 1996; Rhee, 1996). For clarity, we refer to irregular classes as alternating classes and regular classes as non-alternating classes[2].

Alternating conjugation patterns can be further classified into three types depending on the locus of alternation: (1) the stem, (2) the suffix (i.e., the inflectional ending), or (3) both. Within the first type, /p/-, /t/-, and /s/-final stems constitute a particularly interesting case, as their alternations are motivated by the same general assimilatory process—intervocalic lenition.

(1) Intervocalic lenition in alternating classes
  a. p-stems: /p/ → [w] / V_V
  b. t-stems: /t/ → [ɾ] / V_V
  c. s-stems: /s/ → ∅ / V_V

As (1) illustrates, when vowel-initial suffixes follow alternating-class stems, the stem-final obstruents /p/, /t/, and /s/ undergo lenition, surfacing as the sonorants [w] and [ɾ] or deleting (Lee, 2013; Albright and Kang, 2009). The intervocalic environment follows from Korean phonotactics, which disallow consonant clusters, requiring these consonants to be preceded by a vowel. Because this process is lexically restricted, stems belonging to the non-alternating class remain invariant in the same environment when followed by a vowel-initial suffix. This contrast is illustrated in Table 1.

Class	Stem	Conjugation
p-alternating	/ka.ka**p**/ 'close'	[ka.ka.**w**-ʌ]
p-non-alternating	/i**p**/ 'wear'	[i.**p**-ʌ]
t-alternating	/kjɛ.da**t**/ 'realize'	[kjɛ.da.**ɾ**-a.jo]
t-non-alternating	/mu**t**/ 'bury'	[mu.**t**-ʌ]
s-alternating	/na**s**/ 'better'	[na.-a.do]
s-non-alternating	/u**s**/ 'laugh'	[u.**s**-ʌ]

Table 1: Alternating vs Non-alternating classes in V_V

Meanwhile, when stems are followed by consonant-initial suffixes, the environment triggering alternation is absent. Consequently, contrast between classes observed in Table 1 is neutralized: the underlying /p/, /t/ and /s/[3] surface faithfully in both classes. As shown in Table 2, both alternating and non-alternating stems preserve their underlying codas before the declarative suffix *-ta*.

To summarize, the phenomenon exhibits three

[1] Our experimental code and data are available at https://github.com/starry-eyed-s/scaled_conjugation.

[2] This traditional distinction is largely stipulative, as the forms labeled *irregular* are precisely those undergoing phonologically motivated processes and in fact outnumber the regular forms.

[3] When /s/-final stems are followed by consonants, the coda surfaces as [t] due to an independent coda neutralization rule, mapping all obstruents to their homorganic lenis stop in syllable-final position (Jun, 2010). This rule is not relevant to the alternation under investigation and is therefore set aside in the present analysis.

key properties. First, it shows lexical idiosyncrasy: only alternating (irregular) stems undergo lenition. Second, it involves paradigm-specific repair, whereby each consonant-final alternating class adopts a distinct lenition strategy to avoid the *V-OBS-V configuration. Third, the process reflects stem-based exceptionality, with suffixes merely providing the phonological environment that triggers the alternation (intervocalic with -V suffixes and non-intervocalic with -CV suffixes). Hereafter, we refer to suffixes as *environmental triggers*.

Class	Stem	Conjugation
p-alternating	/ka.ka**p**/ 'close'	[ka.ka**p**.-ta]
p-non-alternating	/i**p**/ 'wear'	[i**p**.-ta]
t-alternating	/kjɛ.da**t**/ 'realize'	[kjɛ.da**t**.-ta]
t-non-alternating	/mu**t**/ 'bury'	[mu**t**.-ta]
s-alternating	/na**s**/ 'better'	[na**s**.-ta]
s-non-alternating	/u**s**/ 'laugh'	[u**s**.-ta]

Table 2: Alternating vs Non-alternating classes in V_C

3 Previous approaches towards learning lexical specificity

3.1 The lexical indexation approach

Among the two widely discussed approaches to lexical exceptionality—cophonology and lexical indexation—Korean conjugation has been analyzed within the framework of lexical indexation. Lee (2013) adopts the Recursive Constraint Demotion (RCD) algorithm (Tesar and Smolensky, 1998; Tesar and Prince, 2003) in conjunction with constraint cloning (Pater, 2009) to address the ranking conflict between alternating and non-alternating classes within each paradigm.

RCD incrementally constructs a constraint hierarchy by installing the best-performing constraint—one that prefers only winning candidates—immediately below the established ranking. However, in cases of lexical exceptionality, no single constraint ordering satisfies all winner–loser comparisons: ranking *V-OBS-V below the faithfulness constraint correctly favors the non-alternating-class winner but incorrectly prefers the losing candidate for the alternating class, whereas ranking *V-OBS-V above the faithfulness constraint produces the opposite result.

The conflicting requirements for alternating stems (*V-OBS-V ≫ FAITH) and non-alternating stems (FAITH ≫ *V-OBS-V) are resolved by cloning *V-OBS-V and indexing its applicability separately to the two classes. This yields the ranking scheme in (1), where both orderings are satisfied through a partial ordering of class-specific constraints.

$$*\text{V-OBS-V}_{\text{Alter}} \gg \text{FAITH} \gg *\text{V-OBS-V}_{\text{Non-Alter}} \tag{1}$$

What remains unclear is how separate instantiations of (1) for individual paradigms collectively account for the grammar as a whole. Because candidates in the /p/-, /t/-, and /s/-paradigms adopt different repair strategies and violate distinct faithfulness constraints, competition arises among the faithfulness constraints occupying the ranking space between *V-OBS-V$_{\text{Alter}}$ and *V-OBS-V$_{\text{Non-Alter}}$.

In the /p/-paradigm, the relevant faithfulness constraint violated by repair to [w] is IDENT(CONSONANTAL); in the /t/-paradigm, repair to [ɾ] violates IDENT(LATERAL); and in the /s/-paradigm, deletion of the obstruent segment violates MAX-C. Consequently, in each paradigm the faithfulness constraint implicated in the repair must be outranked by the corresponding faithfulness constraints active in the other two paradigms. To capture these cross-repair ranking relations, the ranking scheme in (1) must be expanded as in (2).

$$\begin{aligned}
&*\text{V-OBS-V}_{\text{P-Alter}} \gg \text{IDENT(LATERAL)}, \text{MAX-C} \gg \\
&\quad \text{IDENT(CONSONANTAL)} \gg *\text{V-OBS-V}_{\text{P-Non-Alter}} \\
&*\text{V-OBS-V}_{\text{T-Alter}} \gg \text{IDENT(CONSONANTAL)}, \text{MAX-C} \gg \\
&\quad \text{IDENT(LATERAL)} \gg *\text{V-OBS-V}_{\text{T-Non-Alter}} \\
&*\text{V-OBS-V}_{\text{S-Alter}} \gg \text{IDENT(CONSONANTAL)}, \\
&\quad \text{IDENT(LATERAL)} \gg \text{MAX-C} \gg *\text{V-OBS-V}_{\text{S-Non-Alter}}
\end{aligned} \tag{2}$$

Undesirably, the expanded ranking requires the learner to maintain paradigm-specific instantiations of the grammar in addition to indexed constraints. This results in a learning complexity that inherits the burdens of both lexical indexation and the cophonology approach: the learner must track both paradigm-specific rankings and the application domains of cloned constraints.

3.2 Lexically Scaled MaxEnt

To address this issue, this paper adopts the lexically scaled MaxEnt model of Linzen et al. (2013); Hughto et al. (2019) as a unified explanatory framework that captures paradigm-specific patterns without positing separate grammars.

$$\mathcal{H}_i = \sum_{\gamma \in \Gamma} \left(w_\gamma + \sum_{m \in \mu_i} s_{\gamma m} \right) v_{\gamma i} \tag{3}$$

The model is a lexicalized variant of the Maximum Entropy (MaxEnt) grammar (Goldwater and Johnson, 2003), motivated to account for languages that exhibit both variability and lexical exceptionality. While variability is handled by the MaxEnt probability assigned by the learner, exceptionality is captured by modifying the harmony score used to compute that probability to incorporate morpheme-specific constraint weights. In (3), morpheme-specific scaling factors $s_{\gamma m}$ are added to the general constraint weight w_γ before being multiplied by the violation profile of candidate i, $v_{\gamma i}$. As indicated by the subscripts, the general weight w_γ applies grammar-wide, whereas the scaling factors $s_{\gamma m}$ are specific to the individual morphemes m that constitute a given candidate.

This additive penalty is conceptually analogous to lexical indexation: a higher-ranked class-specific markedness constraint (e.g., *V-OBS-V$_{\text{Alter}}$ in (1)) can be recast as an increased scaling factor applying to that lexical class. To illustrate this correspondence, consider the idealized learning outcome for the present phenomena shown in Table 3.

	*V-OBS-V	MAX	ID-LAT	ID-C
General weights	4	6	6	6
p-alternating	6	3	2	0
p-non-alternating	0	1	1	1
t-alternating	6	3	0	4
t-non-alternating	0	1	1	1
s-alternating	6	0	3	3
s-non-alternating	0	1	1	1
suffixes	0	0	0	0

Table 3: Hypothetical Learning Result

The rows beneath general weights indicate scaling factors associated with each stem class. Because *V-OBS-V must be ranked above the faithfulness constraints in alternating classes but not in non-alternating classes, only the former bear positive scale weights on *V-OBS-V. As larger weights translate into higher rankings, alternating classes achieve *V-OBS-V$_{\text{Alter}}$ $\gg$ FAITH, whereas the non-alternating classes, receiving no scale weight, obey the ordering FAITH $\gg$ *V-OBS-V$_{\text{Non-Alter}}$. The two partial orders correspond to (2).

Among the scaling factors for the faithfulness constraints, the lowest weight is assigned to the constraint violated by the paradigm-specific alternating class: ID-C for the /p/-alternation, ID-LAT for the /t/-alternation, and MAX for the /s/-alternation[4]. This produces the effect of cross-repair competition within each paradigm.

Finally, suffixes that function as *environmental triggers* ideally bear no scale weights, since they are not the locus of exceptionality.

When each conjugated item is evaluated by the learned grammar, the scaling factors associated with its constituent morphemes combine with the general weights to determine the effective constraint ranking. As class-specific scaling factors apply, distinctions along two dimensions—paradigm and exceptionality—are captured while avoiding the ranking complexity posited by the expanded indexation approach in (2).

4 Experiments

4.1 Toy data Generation

Table 4 reproduces the statistics on Korean obstruent-final stems from the Sejong Corpus reported in Lee (2013). The Sejong Corpus, compiled by the National Institute of Korean Language (NIKL) as part of the 21st Century Sejong Project, has been widely used in previous studies (Albright and Kang, 2009; Jun, 2010), making it a well-established source for such statistics.

p-alternating	68 (72.34%)
p-non-alternating	5 (5.32%)
t-alternating	5 (5.32%)
t-non-alternating	7 (7.45%)
s-alternating	5 (5.32%)
s-non-alternating	4 (4.26%)

Table 4: Distributions of p-, t-, and s-final stems in the Sejong Corpus

Following the type counts in Table 4, we generated pseudo-stems with a CV-/p,s,t/ structure, reflecting Korean phonotactics that disallow consonant clusters. Using the Romanization system provided by NIKL[5], onset consonants and vowels were drawn from the Romanized inventory of Korean segments. Onsets were selected from {g, kk, k, d, tt, t, b, pp, p, j, jj, ch, s, ss, h, n, m, r}, while vowels were restricted to the set of monophthongs {a, eo, o, u, eu, i, ae, e, oe, wi}. Random sampling from these sets yielded 180 unique CV sequences. These sequences were assigned without overlap to the six conjugation classes listed above and instantiated

[4]For notational brevity, this modeling work abbreviates IDENT(CONSONANTAL), IDENT(LATERAL), and MAX-C as ID-C, ID-LAT, and MAX.

[5]https://www.korean.go.kr/front_eng/roman/roman_01.do

with the class-appropriate stem-final consonant (/p/, /t/, or /s/). The full list of toy stems is provided in Table 11 in Appendix A.

Because alternation is triggered in an intervocalic context, the inventory of novel suffixes was balanced with respect to onset presence: suffixes beginning with a consonant (–CV endings) and those beginning with a vowel (–V endings) were equally represented. The initial inventory consisted of two minimal endings, *–ta* and *–a*. It was then incrementally expanded by introducing additional vowel-initial and /t/-initial pairs, with vowels drawn from the same monophthong set used in stem generation. Table 12 in Appendix A summarizes the suffix sets across experimental conditions.

4.2 Distributional Conditions

Table 4 shows that /p/-alternating stems constitute the overwhelming majority of the data (72.34%). Because this distribution creates a substantial statistical imbalance for a stochastic learner, we first conducted experiments under a simplified setting that excluded the /p/-paradigm. The modeling data were therefore restricted to the /t/- and /s/-final stem conjugations, which we refer to as the *balanced condition*. In this setting, we examine two repair strategies for resolving *V-OBS-V: deletion and acquisition of the [+lateral] feature. Accordingly, triplets of UR–SR mappings are considered for each input form, corresponding to (a) the faithful candidate, (b) lenition via violation of IDENT(LATERAL), and (c) lenition via violation of MAX-C. Table 13 in Appendix B presents the UR–SR mappings for the four stem classes under the two suffix conditions introduced above (–V and –CV endings).

Subsequently, we extended the model to the full paradigm by incorporating the /p/-ending data, creating the *imbalanced condition*. In this setting, loss of the [+consonantal] feature becomes an additional repair strategy alongside deletion and acquisition of [+lateral]. Accordingly, UR–SR mappings are represented as quadruplets for each input form, corresponding to (a) the faithful candidate, (b) lenition via violation of IDENT(CONSONANTAL)[6], (c) IDENT(LATERAL), and (d) MAX-C. Table 14 in Appendix B presents a subset of the input tableaux under the two-suffix condition.

[6]When constructing candidate (b) for the -/t/ and -/s/ paradigms, the segmental alternation was instantiated as [j], a glide that preserves the place of articulation, similar to how /p/ surfaces as [w] while preserving labiality.

4.3 Model Parameters

Learner settings followed the default implementation in Hughto et al. (2019). We used an L1 prior ($\lambda = 1.0$) and optimized the model using gradient descent with clipping (Tsuruoka et al., 2009). Weights and scales were initialized to 0.0, and training proceeded for 20,000 epochs with a learning rate of 0.001. All parameters were held constant across experimental conditions.

5 Results

5.1 Learning with the -/t/ and -/s/ paradigms

When only -/t/- and -/s/-final stems were included, type frequencies were roughly balanced both across paradigms and within each paradigm (Table 4). Nevertheless, the attested forms exhibit a 1:3 alternation ratio, because intervocalic lenition occurs only when alternating stems are followed by vowel-initial suffixes; in the other three configurations—alternating stem + CV suffix, non-alternating stem + V suffix, and non-alternating stem + CV suffix—faithful realization prevails. Given this overall distribution, the learner assigned less general weight to *V-OBS-V than to IDENT(LATERAL) and MAX-C.

	*V-OBS-V	MAX	ID-LAT
General weights	0	0.79	0.79
t-alternating stems	0	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	2.16	2.16

Table 5: Learned weights (/t, s/ stems; two suffixes)

As such, while the majority pattern—faithful surfacing—is captured through general weights, the alternations are handled through lexical scaling. This division reflects the model's *least-cost* principle, which prioritizes general mechanisms for capturing dominant patterns while reserving lexical mechanisms for more restricted ones. The same pressure toward economy also shapes the learning of scaling factors. Jarosz (2025) shows that lexicalization tends to be stronger for higher-frequency items than for lower-frequency ones, and this tendency manifests in scale weights initially being mistakenly attributed to the non-idiosyncratic suffixes.

When only two suffix types are available (Table 5), each stem occurs only twice in the attested forms—once with *-ta* and once with *-a*—while

these two suffixes combine with all stem types. Consequently, when frequencies are evaluated per morpheme, suffixes occur more often than stems and thus emerge as the more general morpheme selected to account for exceptionality.

The same frequency bias persists within the suffix inventory. Because intervocalic lenition occurs in only one-fourth of the training data, the learner favors the -CV suffix, which creates a non-intervocalic environment, over the -V suffix. The former activates the ranking FAITH ≫ *V-OBS-V, and scaling is therefore assigned to the associated faithfulness constraints rather than to *V-OBS-V of -V suffixes.

	*V-OBS-V	MAX	ID-LAT
General weights	2.2	3.26	3.26
t-alternating stems	0.37	0	0
t-non-alternating stems	0	0	0
s-alternating stems	0.37	0	0
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Table 6: Learned weights (/t, s/ stems; four suffixes)

As the number of suffixes increases, stems combine with a larger set of suffixes and occur more frequently in the attested forms. This increases the likelihood that stem morphemes are lexicalized, reflected in the progressive redistribution of scale weights from suffixes to stems in Tables 6 and 7, and in Tables 15 and 16 in Appendix C.

The redistribution proceeds in stages. First, the learner captures the triggering of alternation by distinguishing alternating and non-alternating classes, assigning scale weight to *V-OBS-V in the former but not in the latter (Table 6). It then learns the paradigm-specific repair, establishing the ranking among faithfulness constraints that compete to derive class-specific forms. From Table 7 onward, the /t/- and /s/-alternating classes assign less scale weight to the violated constraint—ID-LAT and MAX, respectively—than to the competing faithfulness constraint, and this gap widens as additional suffix types are introduced.

This progression is expected under a frequency-sensitive learning mechanism. The /t/- and /s/-alternating classes have identical type frequencies (five types each), so the statistical signal for learning paradigm-specific behavior comes from the corresponding non-alternating classes. These differ only slightly in type frequency (seven for /t/-non-alternating and four for /s/-non-alternating), requiring a larger number of attested forms to detect the distinction. In the present case, four suffix types are minimally required to identify exceptionality and six suffix types to differentiate class-specific alternations.

	*V-OBS-V	MAX	ID-LAT
General weights	1.99	3.66	3.66
t-alternating stems	1.73	0.13	0
t-non-alternating stems	0	0	0
s-alternating stems	1.73	0	0.13
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Table 7: Learned weights (/t, s/ stems; six suffixes)

Improvements in the distribution of scale weights are reflected in the model's prediction accuracy. To evaluate this effect, we examine prediction accuracy for alternating stems in intervocalic contexts, which constitute a statistical minority and are thus harder for the learner to acquire. As shown in Fig. 1, prediction accuracy increases with the number of suffix types[7]. Notably, accuracy exceeds the 50% threshold only in the ten-suffix condition, where the number of suffix types approaches or exceeds the number of stem types per paradigm (12 for /t/-conjugation and 9 for /s/-conjugation).

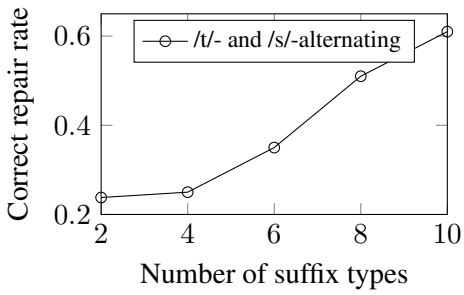

Figure 1: Repair accuracy of /t/- and /s/-alternating stems in intervocalic contexts

5.2 Learning with the -/p/, -/t/, and -/s/ paradigms

Accommodating the /p/-final data creates an imbalance among the three alternating patterns. As shown in Table 4, the /p/-alternating class (72.34%) overwhelmingly outnumbers the /t/-alternating (5.32%) and /s/-alternating classes (5.32%). Although environments that trigger intervocalic lenition are outnumbered by those that block it (approximately in a 1:3 ratio), most lenition tokens

[7]The full numerical results for training accuracy are provided in Table 19 in Appendix D.

come from the /p/-alternating class, such that the number of /p/-alternation tokens (68) approaches the number of faithful realizations (130). Under these conditions, violation of ID-C emerges as the second default pattern after faithful realization. Consequently, when intervocalic lenition is triggered in the -V context, alternating stems favor repair via violation of ID-C.

This skewed distribution directly shapes the model's learning behavior. First, across all suffix conditions (Tables 8–10 and Tables 17 and 18 in Appendix C), the /p/-alternating class never acquires stem-scale weights. Second, ID-C, the constraint violated by this class, shows the greatest resistance to receiving general weight. In the initial two-suffix condition, all of its weight is borne by the *environmental trigger* -CV suffix. This bias persists as the suffix inventory increases and continues to shape the model's learning trajectory.

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	0.69	2.6	2.6	0
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	0
t-alternating stems	0	0	0	0
t-non-alternating stems	0	0	0	0
s-alternating stems	0	0	0	0
s-non-alternating stems	0	0	0	0
-V suffix	0.69	0	0	0
-CV suffix	0	1.92	1.92	4.51

Table 8: Learned Weights (/p, t, s/ stems; two suffixes)

Excluding the effect of ID-C's resistance to receiving general weight, the progression from Tables 8–18 largely parallels the learning results obtained when the /p/ paradigm is excluded. With only two suffixes available, stems initially lack sufficient morphological evidence to learn their indexed status. The second stage—when the model begins to differentiate alternating from non-alternating classes (here limited to /t/ and /s/, since /p/ no longer constitutes the minority pattern captured by lexical scale)—is delayed by the dominance of the /p/-alternations.

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	4.02	5.21	5.21	2.14
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	1.41
t-alternating stems	0	0	0	1.41
t-non-alternating stems	0	0	0	1.41
s-alternating stems	0	0	0	1.41
s-non-alternating stems	0	0	0	1.41
-V suffix	0	0	0	0
-CV suffix	0	0	0	2.15

Table 9: Learned Weights (/p, t, s/ stems; four suffixes)

This is because alternation into a glide functions as a second default pattern alongside faithful surfacing, leaving the /t/- and /s/-alternating classes to compete with two majority patterns. Consequently, even in the four-suffix condition (Table 9), the model fails to differentiate among the six stem classes, as their stored vectors remain identical. Differentiation emerges only in Table 10, where scale weights begin to diverge between the /t/- and /s/-alternating classes.

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	4.98	6.29	6.29	2.71
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	2.54
t-alternating stems	0.61	0	0	2.88
t-non-alternating stems	0	0	0	2.54
s-alternating stems	0.61	0	0	2.88
s-non-alternating stems	0	0	0	2.54
-V suffix	0	0	0	0
-CV suffix	0	0	0	1.53

Table 10: Learned Weights (/p, t, s/ stems; six suffixes)

The effect of statistical imbalance is further reflected in prediction accuracy across the alternating classes. In Fig. 2, the curves for the /t/- and /s/-alternations fail to reach the 50% threshold even when ten suffix types are provided[8], whereas the /p/-alternation maintains substantially higher accuracy throughout training. Nevertheless, the gap between the /p/-paradigm and the /t/- and /s/-paradigms gradually narrows as the number of suffix types increases, again highlighting the role of suffix-type diversity.

6 Discussion

6.1 Lexical statistics

One of the main findings is that, without a sufficiently large number of attested morphemic combinations, generalization cannot develop at the locus of exceptionality. This result highlights the statistical sensitivity of the model when learners are not exposed to enough morphological contexts in which exceptional items may occur.

The results reported for the "lexical language" in Hughto et al. (2019) can be interpreted in the same light. In that study, indexed items (prefixes) were outnumbered by non-indexed ones (stems), and the latter were replicated ten times to achieve the reported performance. By contrast, our experiments were conducted without such frequency

[8] See Table 20 in Appendix D for the exact numerical values.

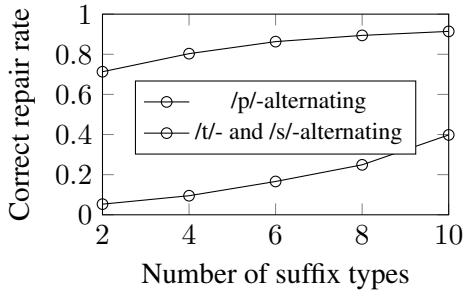

Figure 2: Repair accuracy of /p/-, /t/-, and /s/-alternating stems in intervocalic contexts

asymmetry or token-frequency amplification. Comparing the two results suggests that learning the locus of exceptionality depends critically on sufficient attestation in the input.

This observation raises the question of how suffixes are distributed in natural language. As non-indexed items, suffix statistics have received little attention and lack systematic documentation. We therefore conducted a corpus analysis of the Korean Google Treebank[9]. The corpus shows a distribution of stem types analogous to that in Table 4, while the suffix inventory far exceeds ten, comprising 29 -CV and 33 -V suffixes. Given this diversity in the attested Korean data, absolute type frequency seems unlikely to pose a serious obstacle for real-world language acquisition.

6.2 Empirical Simulation

This finding motivated an additional simulation using the set of conjugations extracted from the corpus to examine whether further learning challenges would arise. In this simulation, we found that although a sufficient number of suffix form is available, stems struggle to learn lexical scales because they are not attested with every suffix type. The difficulty therefore lies not in the absolute number of available types, but in the lack of attested contexts—namely, the sparsity of connections between stems and suffixes.

In our earlier idealized learning scenario, full paradigm saturation was assumed, whereby every stem combines with all possible suffixes (Chan, 2008; Kodner, 2022). This assumption poorly reflects the sparsity of real input data, which typically follows a Zipfian distribution: a small number of stems appear in all derivational forms, while most combine with just a few suffixes. Our empirical simulation likewise reflected this skewness. Only a small subset of stems received scale weights, and among those, few learned sizable values.

By contrast, sizable scaling factors were assigned to suffixes, likely reflecting their higher connectivity with stems. While most stems are attested with only a few suffixes, suffixes themselves occur widely among stems. As a result, stems struggle to acquire scaling factors despite the availability of diverse suffix types. This situation parallels the earlier two-suffix condition, in which stems were attested only twice; here, however, the challenge arises not from a limited number of morphemic counterparts but from sparse stem–suffix connectivity.

6.3 Language acquisition

While analytically unexpected, the initial two-suffix condition—where lexical idiosyncrasy is attributed to suffixes—can be interpreted as a phase of overregularization. Because suffixes provide contextual information, the learner may initially place excessive weight on their behavior, associating alternation with a general phonological rule before learning the locus of exceptionality itself.

This interpretation complements the findings of Jarosz (2025), who show that, unlike in human language acquisition, regularization in the model strengthens over the course of training.[10] Increasing the suffix inventory approximates a learning scenario where the learner is exposed to progressively richer input. In this context, the early two-suffix condition may correspond to an overregularization-like phase in the model's learning trajectory, suggesting that the model's behavior may not entirely diverge from patterns observed in human acquisition.

Nevertheless, two gaps remain if the model is to fully account for the human learning of paradigm-specific exceptionality. First, it must address how appropriate scaling factors can be learned under conditions of sparse stem–suffix connectivity. Second, it must account for how minority phonological processes—such as those in the /t/ and /s/ paradigms—can be learned correctly in the presence of a dominant pattern.

[9] https://github.com/emorynlp/ud-korean/tree/master/google. The stem and suffix inventories identified in the corpus are reported in Tables 22 and 23 in Appendix E

[10] Their time-course analysis tracks this tendency within a single training setting, whereas our comparison spans different suffix conditions. Nevertheless, because all simulations use batch learning and run for the same number of epochs, the results are comparable.

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A Appendix

Category	Stems
p-alternating	neop- ppaep- sep- ssep- gop- bwip- seup- dep- dop- baep- nep- kkeup- moep- kkip- kkop- ssaep- chap- rep- kop- teup- cheop- tep- heup- rwip- jjwip- paep- bop- bup- mip- dwip- choep- sop- swip- pwip- jjeop- keop- soep- ssap- taep- mwip- rup- chwip- bep- peup- ppep- chop- keup- geup- nip- ppip- saep- noep- kkap- jjup- gwip- jjep- meup- ttop- jep- daep- joep- mop- mep- ssoep- ttaep- reop- jjop- ppap-
p-non-alternating	jaep- kkep- jjip- rap- toep-
t-alternating	tat- jat- ket- pot- chit-
t-non-alternating	ppeut- goet- jot- pput- nat- kkut- sit-
s-alternating	ges- kkaes- hoes- koes- nos-
s-non-alternating	poes- meos- jjaes- ches-

Table 11: Toy stems

Condition	Suffix inventory
2-suffix	*-ta*, *-a*
4-suffix	*-ta*, *-a*, *-te*, *-e*
6-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*
8-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*, *-to*, *-o*
10-suffix	*-ta*, *-a*, *-te*, *-e*, *-tu*, *-u*, *-to*, *-o*, *-ti*, *-i*

Table 12: Toy suffixes

B Appendix

UR	Candidate	% occurrences	*V-OBS-V	MAX	ID-LAT
/tat-a$_{\text{T-Alter}}$/	(a) [tata]	0.0	1	0	0
	(b) [tara]	1.0	0	0	1
	(c) [taa]	0.0	0	1	0
/tat-ta$_{\text{T-Alter}}$/	(a) [tatta]	1.0	0	0	0
	(b) [tarta]	0.0	0	0	1
	(c) [tata]	0.0	0	1	0
/jot-a$_{\text{T-Non-Alter}}$/	(a) [jota]	1.0	1	0	0
	(b) [jora]	0.0	0	0	1
	(c) [joa]	0.0	0	1	0
/jot-ta$_{\text{T-Non-Alter}}$/	(a) [jotta]	1.0	0	0	0
	(b) [jorta]	0.0	0	0	1
	(c) [jota]	0.0	0	1	0
/ges-a$_{\text{S-Alter}}$/	(a) [gesa]	0.0	1	0	0
	(b) [gera]	0.0	0	0	1
	(c) [gea]	1.0	0	1	0
/ges-ta$_{\text{S-Alter}}$/	(a) [gesta]	1.0	0	0	0
	(b) [gerta]	0.0	0	0	1
	(c) [geta]	0.0	0	1	0
/poes-a$_{\text{S-Non-Alter}}$/	(a) [poesa]	1.0	1	0	0
	(b) [poera]	0.0	0	0	1
	(c) [poea]	0.0	0	1	0
/poes-ta$_{\text{S-Non-Alter}}$/	(a) [poesta]	1.0	0	0	0
	(b) [poerta]	0.0	0	0	1
	(c) [poeta]	0.0	0	1	0

Table 13: Input data tableau for a statistically-balanced condition

UR	Candidate	% occurrences	*V-OBS-V	MAX	ID-C	ID-LAT
/neop-a$_{\text{P-Alter}}$/	(a) [neopa]	0.0	1	0	0	0
	(b) [neowa]	1.0	0	0	1	0
	(c) [neola]	0.0	0	0	0	1
	(d) [neoa]	0.0	0	1	0	0
/neop-ta$_{\text{P-Alter}}$/	(a) [neopta]	1.0	1	0	0	0
	(b) [neowta]	0.0	0	0	1	0
	(c) [neolta]	0.0	0	0	0	1
	(d) [neota]	0.0	0	1	0	0
/jaep-a$_{\text{P-Non-Alter}}$/	(a) [jaepa]	1.0	1	0	0	0
	(b) [jaewa]	0.0	0	0	1	0
	(c) [jaela]	0.0	0	0	0	1
	(d) [jaea]	0.0	0	1	0	0
/jaep-ta$_{\text{P-Non-Alter}}$/	(a) [jaepta]	1.0	1	0	0	0
	(b) [jaewta]	0.0	0	0	1	0
	(c) [jaelta]	0.0	0	0	0	1
	(d) [jaeta]	0.0	0	1	0	0
/tat-a$_{\text{T-Alter}}$/	(a) [tata]	0.0	1	0	0	0
	(b) [taja]	0.0	0	0	1	0
	(c) [tara]	1.0	0	0	0	1
	(d) [taa]	0.0	0	1	0	0
/tat-ta$_{\text{T-Alter}}$/	(a) [tatta]	1.0	0	0	0	0
	(b) [tajta]	0.0	0	0	1	0
	(c) [tarta]	0.0	0	0	0	1
	(d) [tata]	0.0	0	1	0	0
/jot-a$_{\text{T-Non-Alter}}$/	(a) [jota]	1.0	1	0	0	0
	(b) [joja]	0.0	0	0	1	0
	(c) [jora]	0.0	0	0	0	1
	(d) [joa]	0.0	0	1	0	0
/jot-ta$_{\text{T-Non-Alter}}$/	(a) [jotta]	1.0	0	0	0	0
	(b) [jojta]	0.0	0	0	1	0
	(c) [jorta]	0.0	0	0	0	1
	(d) [jota]	0.0	0	1	0	0
/ges-a$_{\text{S-Alter}}$/	(a) [gesa]	0.0	1	0	0	0
	(b) [geja]	0.0	0	0	1	0
	(c) [gera]	0.0	0	0	0	1
	(d) [gea]	1.0	0	1	0	0
/ges-ta$_{\text{S-Alter}}$/	(a) [gesta]	1.0	0	0	0	0
	(b) [gejta]	0.0	0	0	1	0
	(c) [gerta]	0.0	0	0	0	1
	(d) [geta]	0.0	0	1	0	0
/poes-a$_{\text{S-Non-Alter}}$/	(a) [poesa]	1.0	1	0	0	0
	(b) [poeja]	0.0	0	0	1	0
	(c) [poera]	0.0	0	0	0	1
	(d) [poea]	0.0	0	1	0	0
/poes-ta$_{\text{S-Non-Alter}}$/	(a) [poesta]	1.0	0	0	0	0
	(b) [poejta]	0.0	0	0	1	0
	(c) [poerta]	0.0	0	0	0	1
	(d) [poeta]	0.0	0	1	0	0

Table 14: Input data tableau for statistically imbalanced condition

C Appendix

	*V-OBS-V	MAX	ID-LAT
General weights	1.80	3.85	3.85
t-alternating stems	2.76	0.75	0
t-non-alternating stems	0	0	0
s-alternating stems	2.76	0	0.75
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Table 15: Learned weights (/t, s/ stems; eight suffixes)

	*V-OBS-V	MAX	ID-LAT
General weights	1.72	4.04	4.04
t-alternating stems	3.43	1.14	0
t-non-alternating stems	0	0	0
s-alternating stems	3.43	0	1.14
s-non-alternating stems	0	0	0
-V suffix	0	0	0
-CV suffix	0	0	0

Table 16: Learned weights (/t, s/ stems; ten suffixes)

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	6.14	7.76	7.76	3.67
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.24
t-alternating stems	1.62	0.0005	0	4.09
t-non-alternating stems	0	0	0	3.24
s-alternating stems	1.62	0	0.0005	4.09
s-non-alternating stems	0	0	0	3.24
-V suffix	0	0	0	0
-CV suffix	0	0	0	0.58

Table 17: Learned Weights (/p, t, s/ stems; eight suffixes)

	*V-OBS-V	MAX	ID-LAT	ID-C
General Weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

Table 18: Learned Weights (/p, t, s/ stems; ten suffixes)

D Appendix

Two suffixes

	t-conjugation				s-conjugation			
	t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.523	0.904	0.523	0.904	0.523	0.904	0.523	0.904
P(lateralization)	**0.238**	0.047	0.238	0.047	0.238	0.047	0.238	0.047
P(deletion)	0.238	0.047	0.238	0.047	**0.238**	0.047	0.238	0.047

Four suffixes

	t-conjugation				s-conjugation			
	t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.50	0.93	0.59	0.93	0.50	0.93	0.59	0.93
P(lateralization)	**0.25**	0.03	0.20	0.03	0.25	0.03	0.20	0.03
P(deletion)	0.25	0.03	0.20	0.03	**0.25**	0.03	0.20	0.03

Six suffixes

	t-conjugation				s-conjugation			
	t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.33	0.95	0.73	0.95	0.33	0.95	0.73	0.95
P(lateralization)	**0.35**	0.024	0.14	0.03	0.31	0.024	0.14	0.02
P(deletion)	0.31	0.021	0.14	0.021	**0.35**	0.024	0.14	0.02

Eight suffixes

	t-conjugation				s-conjugation			
	t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.25	0.97	0.79	0.96	0.25	0.97	0.79	0.96
P(lateralization)	**0.51**	0.10	0.21	0.02	0.24	0.01	0.10	0.02
P(deletion)	0.24	0.01	0.09	0.02	**0.51**	0.02	0.10	0.02

Ten suffixes

	t-conjugation				s-conjugation			
	t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.20	0.97	0.84	0.96	0.20	0.987	0.84	0.96
P(lateralization)	**0.61**	0.02	0.08	0.02	0.19	0.003	0.08	0.02
P(deletion)	0.19	0.01	0.08	0.02	**0.61**	0.01	0.08	0.02

Table 19: Training-data P(E) under the balanced condition

Two suffixes

	p-conjugation				t-conjugation				s-conjugation			
	p-alternating		p-non-alternating		t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.181	0.968	0.181	0.968	0.181	0.968	0.181	0.968	0.181	0.968	0.181	0.968
P(glide)	**0.713**	0.011	0.713	0.011	0.713	0.011	0.713	0.011	0.713	0.011	0.713	0.011
P(lateralization)	0.053	0.011	0.053	0.011	**0.053**	0.011	0.053	0.011	0.053	0.011	0.053	0.011
P(deletion)	0.053	0.011	0.053	0.011	0.053	0.011	0.053	0.011	**0.053**	0.011	0.053	0.011

Four suffixes

	p-conjugation				t-conjugation				s-conjugation			
	p-alternating		p-non-alternating		t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.123	0.976	0.313	0.986	0.313	0.986	0.313	0.986	0.313	0.986	0.313	0.986
P(glide)	**0.803**	0.013	0.497	0.003	0.497	0.003	0.497	0.003	0.497	0.003	0.497	0.003
P(lateralization)	0.037	0.005	0.095	0.005	**0.095**	0.005	0.095	0.005	0.095	0.005	0.095	0.005
P(deletion)	0.037	0.005	0.095	0.005	0.095	0.005	0.095	0.005	**0.095**	0.005	0.095	0.005

Six suffixes

	p-conjugation				t-conjugation				s-conjugation			
	p-alternating		p-non-alternating		t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.089	0.982	0.434	0.995	0.334	0.995	0.434	0.995	0.334	0.995	0.434	0.995
P(glide)	**0.863**	0.014	0.332	0.001	0.333	0.001	0.332	0.001	0.333	0.001	0.332	0.001
P(lateralization)	0.024	0.002	0.117	0.002	**0.166**	0.002	0.117	0.002	0.166	0.002	0.117	0.002
P(deletion)	0.024	0.002	0.117	0.002	0.166	0.002	0.117	0.002	**0.166**	0.002	0.117	0.002

Eight suffixes

	p-conjugation				t-conjugation				s-conjugation			
	p-alternating		p-non-alternating		t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.076	0.985	0.538	0.998	0.25	0.998	0.538	0.998	0.25	0.998	0.537	0.998
P(glide)	**0.894**	0.014	0.249	0.0005	0.25	0.0002	0.249	0.0005	0.25	0.0002	0.249	0.0005
P(lateralization)	0.015	0.0004	0.106	0.0004	**0.249**	0.0004	0.106	0.0004	0.249	0.0004	0.106	0.0004
P(deletion)	0.015	0.0004	0.106	0.0004	0.249	0.0004	0.106	0.0004	**0.249**	0.0004	0.106	0.0004

Ten suffixes

	p-conjugation				t-conjugation				s-conjugation			
	p-alternating		p-non-alternating		t-alternating		t-non-alternating		s-alternating		s-non-alternating	
	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV	-V	-CV
P(faithful)	0.064	0.988	0.61	0.999	0.2	0.999	0.61	0.999	0.2	0.999	0.611	0.999
P(glide)	**0.914**	0.011	0.20	0.0002	0.2	0	0.2	0	0.2	0	0.2	0
P(lateralization)	0.010	0.0001	0.094	0.0001	**0.398**	0	0.094	0	0.2	0	0.094	0
P(deletion)	0.010	0.0001	0.094	0.0001	0.2	0	0.09	0	**0.398**	0.0001	0.09	0

Table 20: Training-data P(E) under the imbalanced condition

E Appendix

p-alternating 45 (45%)
p-non-alternating 17 (17%)
t-alternating 8 (8%)
t-non-alternating 14 (14%)
s-alternating 8 (8%)
s-non-alternating 8 (8%)

Table 21: Distribution of /p/, /t/, and /s/-final stems in the Google Treebank Corpus

Category	Stems
p-alternating	gakkap- gomap- gop- gup- geurip- deop- dop- dukkeop- tteugeop- maep- mugeop- museop- budeureop- saerop- swip- sikkeureop- akkap- areumdap- aswip- antakkap- eoryeop- jeulgeop- chup- himgyeop- kkadarop- deoreop- maekkeureop- bangap- bukkeureop- sonswip- singgeop- useup- jeonggyeop- gabyeop- goerop- gwiyeop- nollap- nup- duryeop- duteop- maeseop- ansseureop- eodup- oerop- chagap-
p-non-alternating	dwijip- butjap- ppob- ssip- ip- jap- job- barojap- hwijip- kkojip- kkob- sarojap- sonjap- jip- jjip- charyeip- himip-
t-alternating	geot- deut- mut- sit- kkaedat- ilkkeut- git- but-
t-non-alternating	dat- mit- bat- ssot- eot- gut- mulryeobat- beorimbat- injeongbat- geot- got- tteut- mut- ieobat-
s-alternating	nas- bus- is- jis- maedeupjis- nongsajis- dwis- yeonis-
s-non-alternating	ssis- us- johas- chisos- beos- byas- ppaeas- is-

Table 22: /p/, /t/, and /s/-final stems in the Google Treebank Corpus (romanized)[a]

[a] Romanization follows https://www.korean.go.kr/front_eng/roman/roman_01.do.

Context	Suffix inventory
-CV	-geona -ge -get -go -goyo -guyo -gi -gillae -na -nayo -neyo -neureago -neun -neunda -neundago -neunde -da -daga -dago -daneun -deogunyo -deon -dorok -seumnida -ja -janeun -jamaja -ji -jiman
-V	-n -nde -l -m -myeo -myeon -myeonse -a -ado -aseo -aya -ayo -at -eo -eodo -eoseo -eoya -eoyo -eot -eotseot -euna -euni -eureo -eureogo -eureoneun -eumyeo -eumyeon -eumyeonse -euseyo -eusi -eun -eul -eum

Table 23: Suffix inventories in the Google Treebank Corpus (romanized)[a]

[a] Romanization follows https://www.korean.go.kr/front_eng/roman/roman_01.do.